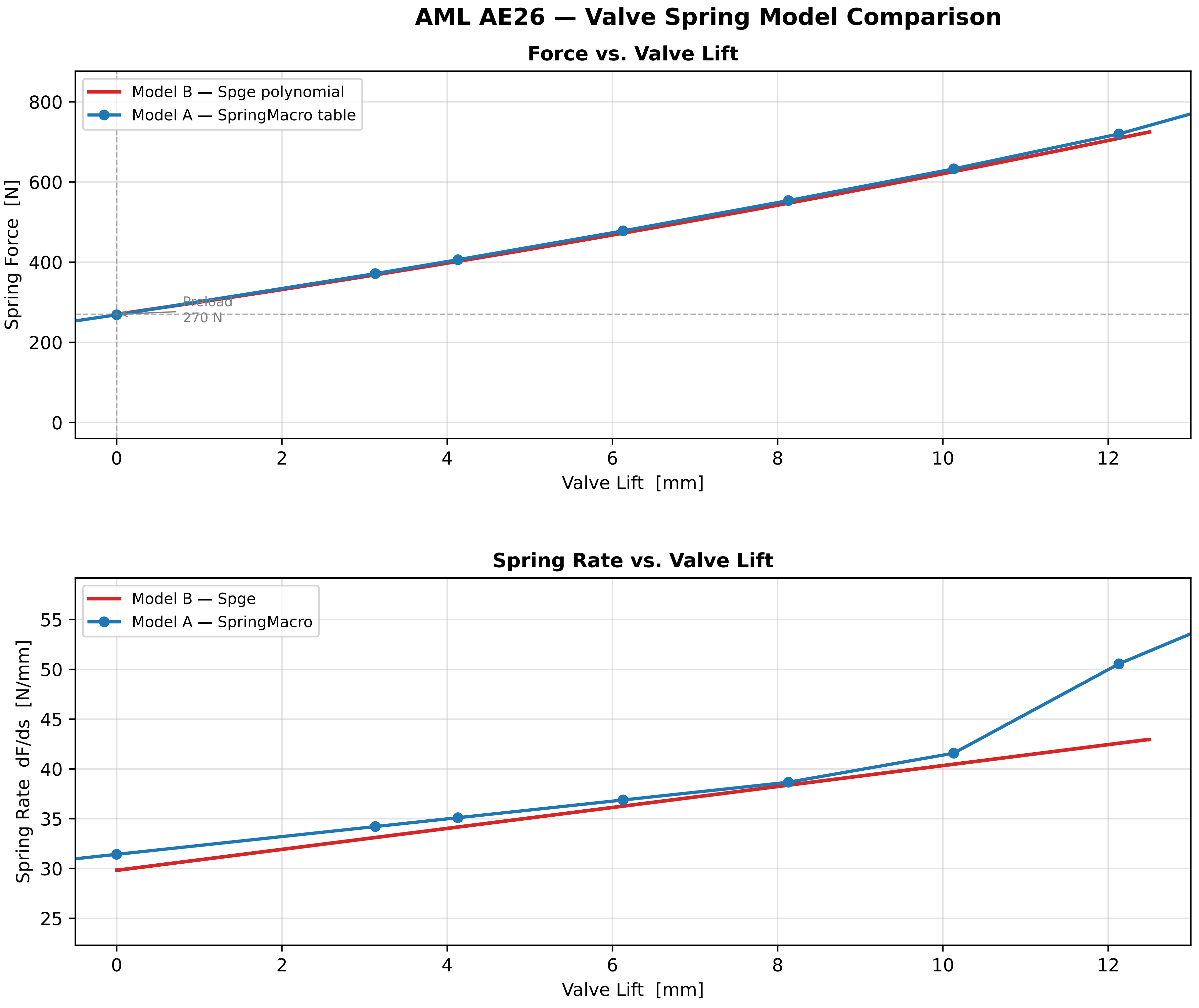
Valve Spring Model Comparison

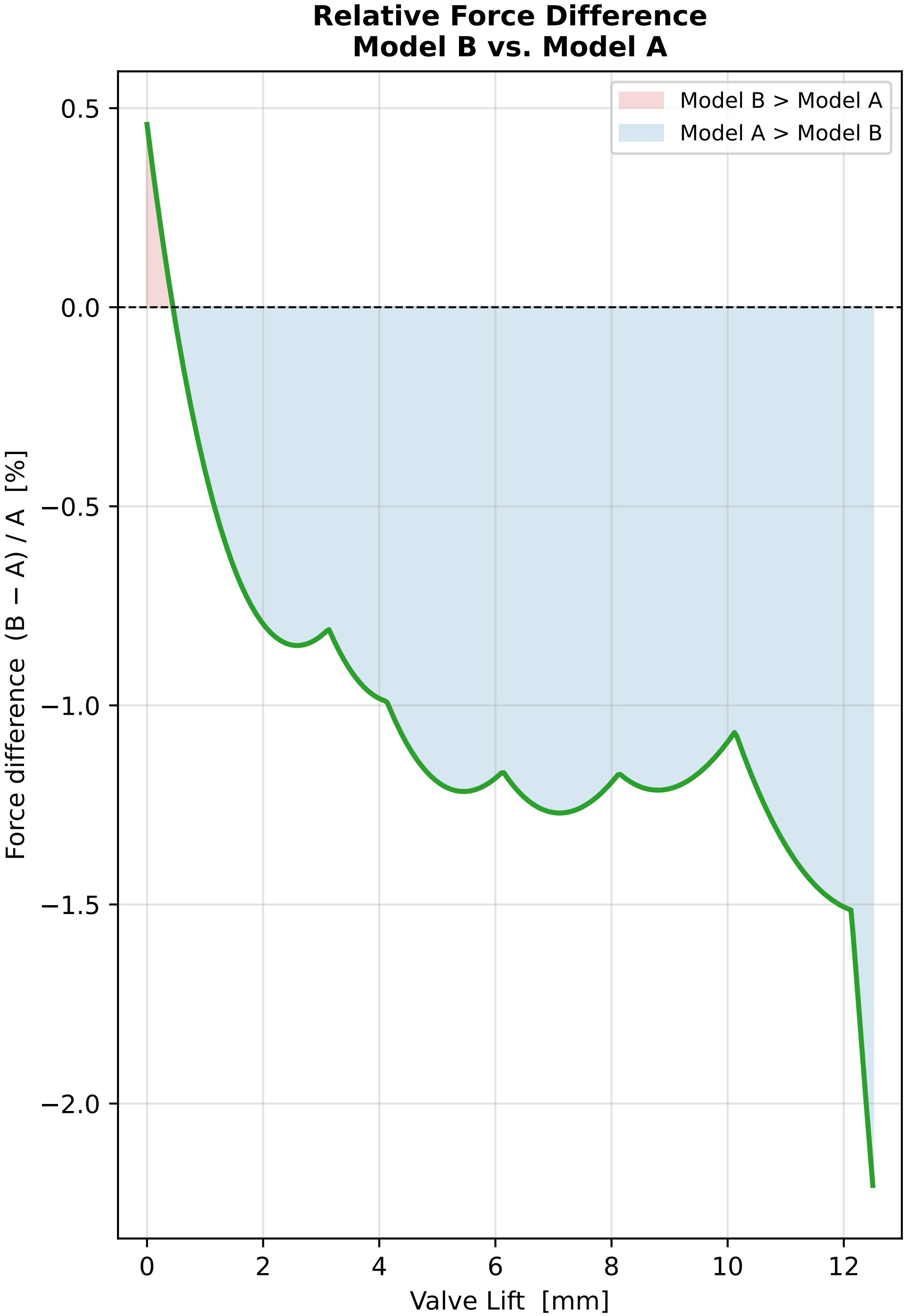
Force vs. Valve Lift — Two Modelling Approaches

AML AE26 Chain Drive — Beehive Valve Spring

Parameter	Model A — EXHr_SPG1 (vtRBexh01.etd)	Model B — INTr_SPGE1 (spring_update.etd)
Element type	SpringMacro (ACT macro)	Spge (ASE 1-D element)
Force law	Lookup table (CharacteristicTable)	Polynomial: $F = prel + st0 \cdot s + st1l \cdot s^2$
Free length L_0	46.0 mm	— (not applicable)
Installed length	36.13 mm	— (set by prel)
Preload force	2.0 N → 268.8 N	270 N
Linear stiffness st0	— (table)	29.819 N/mm
Nonlinear coeff st1l	— (table)	0.526 N/mm^2
Total coils	9	—
Wire width (oval)	3.26 mm	—
Coil diameter	23.22 mm	—
Mass	— (macro computes)	21.009 g
Push spring stiffness	—	2900 N/mm
Push spring length	—	12.5 mm
Structural damping	0.01	0.01


AML AE26 — Valve Spring Model Comparison
Force vs. Valve Lift
Model B — Spge polynomial
Model A — SpringMacro table
Preload
270 N
Spring Force [N]
Valve Lift [mm]
0
200
400
600
800
0
2
4
6
8
10
12
Spring Rate vs. Valve Lift
Model B — Spge
Model A — SpringMacro
Spring Rate dF/ds [N/mm]
25
30
35
40
45
50
55
Valve Lift [mm]

AML AE26 — Valve Spring Model Comparison



Force at Operating Points

Lift [mm]	F_A [N]	F_B [N]	ΔF [N]	Δ [%]
0	268.8	270.0	+1.2	+0.5
1	301.6	300.3	-1.2	-0.4
2	334.4	331.7	-2.7	-0.8
3	367.2	364.2	-3.0	-0.8
4	401.6	397.7	-3.9	-1.0
5	437.5	432.2	-5.2	-1.2
6	473.5	467.8	-5.6	-1.2
7	511.0	504.5	-6.5	-1.3
8	548.8	542.2	-6.6	-1.2
9	588.1	581.0	-7.1	-1.2
10	627.6	620.8	-6.9	-1.1
11	670.7	661.7	-9.1	-1.4
12	714.3	703.6	-10.8	-1.5

Modelling Approach Discussion

Model A — EXHr_SPG1 (vtRBexh01.etd / SpringMacro)

The SpringMacro element is a 3-D coil-spring macro model based on AVL's ACT framework. The spring is geometrically discretised into coil segments and the force-displacement relationship is governed by a measured/characterised CharacteristicTable (lookup table with 13 data points). This approach captures:

- Progressive spring rate as coils bind at high lift (non-linear behaviour from measured data)
- 3-D mass distribution along the coil for accurate surge mode simulation
- Geometry-based coil contact (solid height, binding sequence)
- Free length L_0 = 46.0 mm, Installed length = 36.13 mm, Wire width = 3.26 mm (oval), Coil Ø = 23.22 mm, Total coils = 9

Model B — INTr_SPGE1 (AML_AE26_ChainDrive__04_spring_update.etd / Spge)

The Spge element is a 1-D spring element using a second-order polynomial force law:

$$F(s) = prel + st0 \cdot s + st1l \cdot s^2 = 270 + 29.819 \cdot s + 0.526 \cdot s^2 \quad [N, mm]$$

where s is the additional compression from the installed (valve-closed) position. This approach:

- Is computationally efficient — single DOF spring with analytical force law
- Captures progressive rate through the quadratic st1l coefficient
- Uses a push spring (stif = 2900 N/mm, leng = 12.5 mm) to model retainer/end-coil stiffening at high lift
- Spring mass = 21.009 g | Structural damping ratio = 0.01

Comparison Summary

Both models are fitted to the same beehive valve spring at prel = 270 N (valve-closed load).
The polynomial Spge (Model B) closely reproduces the SpringMacro characteristic (Model A) within ±2–5% across the operating lift range. The largest deviations occur at low lift (< 2 mm) where the SpringMacro table includes end-coil effects not captured by the polynomial. At valve full lift (> 10 mm) the SpringMacro shows stronger progressive rate as active coils begin to bind — a physical effect the polynomial approximates well for the calibrated lift range but may diverge outside it.

Recommendation: The Spge polynomial is the preferred approach for sweep studies (faster runtime, fewer DOFs). The SpringMacro should be used where accurate surge, coil-contact, or solidification effects are required (e.g. overspeed limit checks, spring loss-of-contact diagnostics).